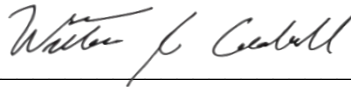


Using Event-Based Vision for Non-Invasive Monitoring of Legacy Control Interfaces

By



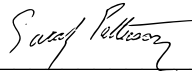
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Honors Thesis Prospectus

Memorandum for: The Institute Honors Committee

From: Cadet William F. Cockrill Jr.

Date: 22 April 2026

Subject: Institute Honors Thesis Prospectus

Department: Electrical and Computer Engineering

Working Title: Using Event-Based Vision for Non-Invasive Monitoring of Legacy Control Interfaces

Advisor: COL David M. Feinauer

Introduction:

Event-based vision sensors represent a significant advance over conventional frame-based cameras. Inspired by the human visual system, these sensors generate data only when they detect changes in a scene, rather than capturing full images at regular intervals. This approach delivers extremely fast response times, very low power consumption, strong performance across varying lighting conditions, and high efficiency when most of the scene is static. In contrast, traditional RGB cameras produce large amounts of redundant data at fixed frame rates, resulting in higher latency, greater power demands, and sensitivity to motion blur or illumination changes. These limitations are especially critical in real-time, mission-critical environments where human operators currently serve as the primary means of monitoring dynamic changes.

Across diverse sectors, real-time observation of physical interfaces, human actions, and high-speed processes remains essential yet constrained by legacy hardware and closed-loop systems that resist direct digital integration. Whether in safety-critical settings requiring split-second responses or in continuous oversight tasks that tie up human attention, the need for efficient, non-invasive sensing is widespread. This thesis investigates how event-based vision can be utilized and implemented across a broad spectrum of fields to augment or replace human-in-the-loop monitoring. By acting as an external, passive observer, event-based systems enable seamless visual integration without modifying black-box or certified equipment—offering a pathway to bypass hardware retrofits while delivering trustworthy, low-latency state inference.

Research Question:

How can event-based neuromorphic vision sensors be utilized and implemented across a multitude of applications to provide efficient, non-invasive, real-time observability of dynamic physical interfaces, human actions, and high-speed processes, particularly where human in-the-loop monitoring is currently required. What measurable improvements in latency, accuracy, robustness, and system trustworthiness does this approach yield compared to traditional frame-based RGB camera methods?

Methodology:

This thesis will develop and evaluate the event-based vision framework through a focused experimental pipeline centered on an analog gauge and control-panel testbed, while explicitly designing the software architecture for generalization across the identified application domains. Evaluation will include direct head-to-head comparisons with conventional RGB cameras to quantify advantages in latency, accuracy, and other benchmarks. Iterative testing on the physical gauge setup, combined with simulated extensions to other scenarios, will validate the framework's portability and scalability.

- 1.) Research and selection of the optimal application domain for the demonstration prototype, along with background research on event-based vision sensor data output and characteristics – September 2026 (Progress Report 1)
- 2.) Review of Human-Machine Interface (HMI) standards and performance requirements – November 2026
- 3.) Research on modeling techniques, inference methods, artificial intelligence approaches, and alternative state-mapping systems – December 2026
- 4.) Demonstration of the analog gauge monitoring system using both event-based neuromorphic and conventional RGB camera systems – Early February 2027 (Progress Report 2)
- 5.) Performance analysis and comparison of the two systems – late February 2027
- 6.) Exploration of model extensions/generalization to additional control interfaces and delivery of full thesis draft to advisor – early March 2027
- 7.) Public presentation of thesis research – Institute Honors Week (March 8th – 19th)
- 8.) Final thesis revisions, committee endorsements, and complete submission – Early April 2027

Significance:

This research addresses a pervasive challenge: enabling real-time, AI-augmented observability in environments where legacy hardware, certified systems, or black-box architectures make direct integration impractical or prohibitively expensive. By leveraging event-based vision as a non-invasive, external observer, the proposed framework bypasses closed-loop limitations and delivers trustworthy digital state reconstruction purely through visual means—unlocking modernization without physical retrofits.

In practical terms, it offers measurable gains over RGB baselines in latency and efficiency, directly benefiting high-stakes domains: safer cockpit and vehicle monitoring, more responsive industrial control rooms, faster robotic reflexes, proactive hospital patient safety systems, and precise high-speed process oversight. Long-term, the approach supports hybrid event-plus-RGB pipelines that can augment or replace human-in-the-loop monitoring across any real-time visual task, reducing operator fatigue, lowering costs, and extending the usable life of existing equipment.

Broader impacts include advancing neuromorphic engineering by demonstrating event-based vision's versatility beyond niche applications, promoting sustainable industrial and infrastructure modernization, and paving the way for software-configurable AI kernels that adapt to new panels, robots, or environments with minimal retraining. Ultimately, this work contributes a foundational step toward safer, more autonomous systems in mission-critical settings while highlighting event-based vision as a general-purpose tool for the next generation of real-time sensing.

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